Today: prove 2 theorems:

- frequency of visits: for $N_n(x) = \#\{m \le n : X_m = x\}$

  if MC is irreducible & recurrent & has stationary distr π,

  almost surely, $\frac{N_n(x)}{n} \xrightarrow{n \to \infty} \pi(x)$

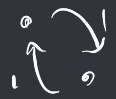
- convergence of trans. probab: if MC irred, recurrent aperiodic & has π,

  $\forall\, x, y, \quad p^n(x,y) \xrightarrow{n \to \infty} \pi(y)$

aperiodicity: period of a state x in a MC is

$$\mathrm{per}(x) := \gcd\{n : p^n(x,x) > 0\}$$

greatest common divisor

ex: in  , both states have $p^n(x,x) = \begin{cases} 0 & n \text{ odd} \\ 1 & n \text{ even} \end{cases}$

so $\{n : p^n(x,x) > 0\} = \{2n : n \in \mathbb{N}\}$

which has gcd $= 2$.

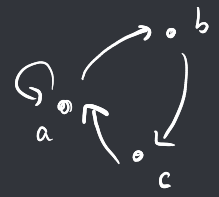
$\therefore$ both states have period 2.

all have period 3.

all have period 2.

if a state x has period > 1, then $p^n(x,x)$ cannot converge (except to 0) as $n \to \infty$.

ex: what is the period?



$$\{n : p^n(b,b) > 0\} = \{3, 4, 5, \dots\}$$

$\text{per}(b) = 1$ since $\gcd(3,4) = 1$.

$$\{n : p^n(a,a) > 0\} = \{2n + 3m : n, m \in \mathbb{N}\}$$
$$= \{2, 3, 4, \dots$$

$\therefore \gcd\{\cdots\} = 1 \quad \text{per}(a) = 1.$

Fact: if $\text{per}(x)=1$ then $\exists\, n_0$: $p^n(x,x)>0$ $\forall n \geq n_0$.

(little bit of number theory: note that $p^{n+m}(x,x) \geq p^n(x,x)\,p^m(x,x)$

so if $p^n(x,x)$, $p^m(x,x)$ both >0

then $p^{n+m}(x,x)>0$).

Frequency of visits.

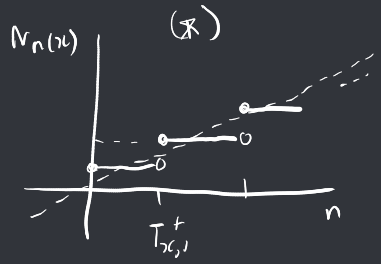
suppose that $X_0 = x$.

If x recurrent, $T_x^+ < \infty$ w.p. 1.

Moreover if $T_{x,k}^+$ = time of kth return to x

then $N_n(x) = k$ iff $T_{x,k-1}^+ \leq n < T_{x,k}^+$

From strong Markov property, excursions from x are i.i.d.



in particular

$$\left(T^+_{x,k} - T^+_{x,k-1} \right)_{k \geq 1}$$

are i.i.d. with mean

$$= \underbrace{E_x\left[T_x^+ \right]}_{<\infty}$$

Law of large numbers: if $< \infty$ then

$$T^+_{x,k} / k \xrightarrow{k \to \infty} E_x\left[T_x^+ \right] \quad \text{almost surely.}$$

The picture suggests that $N_n(x) / n \xrightarrow{n \to \infty} 1 / E_x\left[T_x^+ \right]$

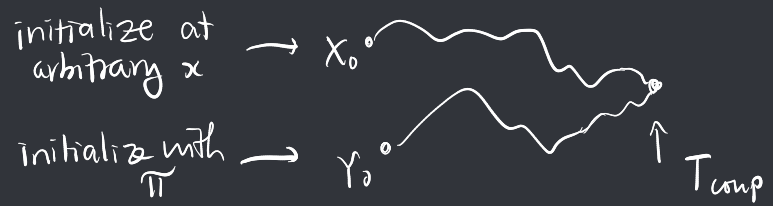
Later we'll show that for any stationary π on irreducible MC,

if $\pi(x) > 0$ then $\pi(x) = \dfrac{1}{E_x[T_x^+]}$.

☆ Convergence of transition probability.

to actually prove this, we will use "coupling".

Idea: construct two copies (X_n, Y_n) of the MC & characterize "distance" between distbtns. by time when they first meet.



Claim: If $P(T_{coup} > n)$ is small, then X_n, Y_n should have "similar" distribution.

Let X_n, Y_n be independent until they meet. After let $X_n = Y_n$. Marginal distbn. of each process remains correct.

☆ Distance between distributions

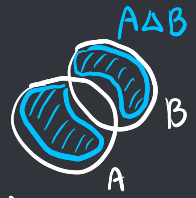
Total variation distance: for probability distributions μ, ν on common space Ω

"mu" "nu"

define $TV(\mu,\nu) = \sup_{A \subset \Omega} |\mu(A) - \nu(A)|$.

For discrete distbtns. $\mu(x), \nu(x)$ ($x \in S$, S finite or countable)

$$TV(\mu,\nu) = \frac{1}{2} \sum_{x \in S} |\mu(x) - \nu(x)|.$$



We can bound TV distance between distbtn. of X_n & Y_n in terms of coupling time:

$$|P(A) - P(B)| \le P(A \Delta B)$$
$$:= P(A \cap B^c) + P(B \cap A^c)$$

$$|P(X_n = y) - P(Y_n = y)| \le P(X_n = y, Y_n \ne y) + P(Y_n = y, X_n \ne y)$$
$$\le P(X_n = y, T_{cup} > n) + P(Y_n = y, T_{cup} > n)$$

by defn $X_n = Y_n \ \forall\, n \ge T_{cup}$

sum over y:
$$\sum_y |P(X_n = y) - P(Y_n = y)| \le P(T_{cup} > n) + P(T_{cup} > n)$$
$$= 2P(T_{cup} > n)$$

In particular if $X_0 = x$ $\quad Y_0 \overset{d}{=} \pi$ then $\quad P(X_n = y) = p^n(x,y)$

& $P(Y_n = y) = \pi(y)$

so

$$|p^n(x,y) - \pi(y)| \leq 2\underbrace{P(T_{\text{comp}} > n)}_{\text{w.t.s this} \to 0 \text{ as } n \to \infty.}$$

ex:

not aperiodic. Why could X_n, Y_n fail to couple?

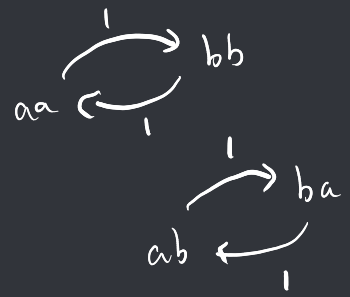
If they start in opposite states

(if $X_0 = a$ or b then $P(Y_0 \neq X_0) = \frac{1}{2}$))

then they never meet almost surely.

The product chain is not irreducible:

two independent copies, state space $S \times S$



This is relevant because $T_{coup} = \inf\{n : (X_n, Y_n) \in D\}$

$D = \{(x,x) : x \in S\} \subset S \times S.$

To make argument work, want product chain irreducible.

This holds if original MC is irred. & aperiodic.

$(x,y) \rightsquigarrow (z,w)$ to see this: need to be able to reach any (z,w) from any (x,y) in some # steps.

Define $\bar{p}$ on $S \times S$: (indep^{dt} copies)

$$\bar{p}((x,y),(u,v)) = p(x,u)\,p(y,v)$$

$$x \xrightarrow{\text{in } k \text{ steps}} z \qquad y \xrightarrow{\text{in } l \text{ steps}} w$$

if irred.

Then for large m,

$$\bar{p}^{k+l+m}((x,y),(z,w))$$

$$\geq p^{k}(x,z)\,p^{l+m}(z,z)\,p^{l}(y,w)\,p^{k+m}(w,w)$$

$$> 0.$$

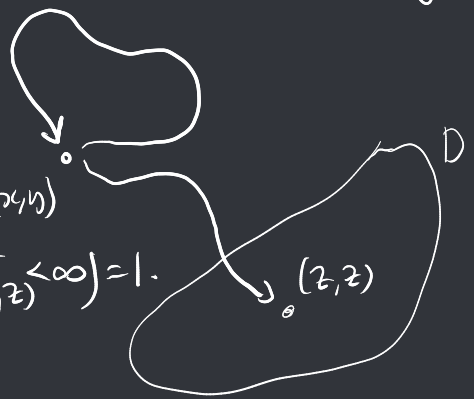
if aperiodic, any large enough # steps

$$z \longrightarrow z$$

$$w \longrightarrow w$$

Last ingredient: recurrence. If product chain irred & recurrent then from any (x,y),

$$P_{(x,y)}(T_{\text{wrp}} > n) \xrightarrow{n\to\infty} 0$$



$$P_{(x,y)}(T^+_{(z,z)} < \infty) = 1.$$

$(T_{\text{wrp}} \le T^+_{(z,z)}$ any given $z)$.

To get recurrence of prod. chain note $\bar{\pi}(x,y) := \pi(x)\pi(y)$

stationary for prod. chain.

& stationary distbn. has zero probab. on transients (done if $\underbrace{\pi(x) > 0 \ \forall x}_{\text{check later for irred.}}$)